

ANALYSTLAB AFRICA

Experience Lab Internship Programme

HealthConnect Clinic — Week 8 Final Report

Final Analytics, Dashboard & Business Insights

Intern Name	Belinda Thando Mhlanga
Track	Data Analytics
Programme	AnalystLab Africa Experience Lab
Project	HealthConnect Clinic
Week	Week 8 — Final Integration & Presentation
Submission Date	27 September 2026
Tools Used	Python · Pandas · Matplotlib · SQL · HTML Dashboard

1. Week 7 → Week 8 Transition Summary

Before entering the final integration stage, the following transition summary documents the outcome of Week 7 testing and the readiness state carried into Week 8.

Transition Item	Detail
Main Week 7 Output	Eight formal KPI tests (all passed) + three analytical refinements + interactive 5-page HTML dashboard
Most Important Test Result	T1–T8: 100% pass rate. KPI 1 confirmed at exactly 48.46% ($2,423 \div 5,000$). All KPI calculations arithmetically verified against two independent methods.
Major Issue Discovered	No data errors or calculation inconsistencies found. The only gap was the absence of a quantified High Risk best-case intervention rate — resolved by Refinement R2.
What Was Refined	R1: December seasonal driver identified (Follow-up + Specialist). R2: High Risk best-case rate established at 37.93% under SMS + same-week booking. R3: Patient waiting time profiled by type and day.
Component Ready	All five KPIs validated. Five-page interactive analytics dashboard complete. Cross-track integration with Data Science confirmed.
Still Needs Integration	Data Analytics findings must feed the Data Science model evaluation framework and Project Management final presentation.
Tracks Involved	Data Analytics (lead) → Data Science (primary consumer) → Project Management (final presentation coordination)
Final Presentation Intent	Demonstrate the complete analytical journey: problem framing → KPI design → EDA → advanced analysis → validation → business recommendations, and show how analytics findings directly shaped the Data Science model.

2. Final Integration Readiness

The following table confirms that the Data Analytics track component is ready for final integration into the HealthConnect solution.

Requirement	Data Analytics Response
What is my final component?	A validated five-KPI analytics framework and interactive five-page decision-support dashboard covering no-show patterns across all key dimensions.
What problem does it address?	HealthConnect Clinic's 48.46% appointment no-show rate — identifying the behavioural, logistical, and temporal factors that predict missed appointments.
What is its current status?	Complete and validated. All 8 formal tests passed in Week 7. Dashboard deployed. Findings delivered to Data Science track.
What did I improve in Week 7?	Confirmed December driver (Follow-up 56.63%, Specialist 55.10%). Established High Risk best-case benchmark (37.93%). Profiled patient waiting time as an operational metric.
Which tracks depend on my output?	Data Science (validated patterns for model feature engineering and evaluation) and Project Management (business insights for final presentation).
Which tracks do I depend on?	Data Science confirmed that the engineered features (lead_time_band, patient_risk_tier) improved model precision and recall — this closed the feedback loop.
What output will I provide?	Final KPI report, validated findings table, interactive dashboard, five business recommendations, and this Week 8 project portfolio.
What evidence demonstrates readiness?	Eight passed tests, two independent method agreement on all age-group rates, 5,000-record completeness check, and zero data integrity violations.
What limitations remain?	No qualitative data on cancellation reasons. Waiting time available for attended patients only. Dataset covers one clinic; findings may not generalise directly.

3. Executive Summary

HealthConnect Clinic faces a critical operational challenge: nearly half of all scheduled appointments result in a missed visit. This project applied data analytics to 5,000 appointment records spanning January 2025 to June 2026 to identify the behavioural, logistical, and temporal factors driving no-shows — and to generate evidence-based recommendations for reducing them.

3.1 The Business Problem

A no-show rate of 48.46% represents empty consulting rooms, wasted clinical capacity, and unserved patients. The challenge is not uniform: certain patient segments, booking windows, and appointment types carry significantly higher risk than others. The central question driving this analysis was:

"How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?"

3.2 Final KPI Dashboard

KPI	Final Validated Result
KPI 1 — Overall No-Show Rate	48.46% (2,423 of 5,000)
KPI 2 — Reminder Effectiveness	4.03 pp reduction (51.39% → 47.36%)
KPI 3 — Lead Time: 0–7 days	27.81% (n = 640)
KPI 3 — Lead Time: 8–14 days	33.55% (n = 602)
KPI 3 — Lead Time: 15–30 days	43.21% (n = 1,333)
KPI 3 — Lead Time: 31–60 days	60.49% (n = 2,425)
KPI 4 — Low Risk patients	43.51% (n = 2,921)
KPI 4 — Medium Risk patients	53.49% (n = 1,548)
KPI 4 — High Risk patients	61.02% (n = 531)
KPI 5 — Follow-up (worst type)	51.23% (n = 1,421)
KPI 5 — General Consultation	46.64% (n = 2,086)
Best segment (LR + 0–7d + SMS)	23.18% (n = 151)
Worst segment (HR + 31–60d + No reminder)	72.31% (n = 65)
Segment gap	49.13 percentage points

3.3 Key Findings

- Booking lead time is the single most powerful predictor of no-shows. Patients who book 31–60 days in advance miss 60.49% of appointments — more than double the 27.81% rate for same-week bookings.

- Patient risk tier compounds lead time. High Risk patients who book 31–60 days ahead with no reminder miss 72.31% of appointments. The same patients given an SMS reminder and a same-week booking miss only 37.93%.
- SMS is the most effective reminder channel across all risk tiers (45.75% no-show rate vs. 49.77% for WhatsApp and 51.39% for no reminder at all).
- December is the worst month (53.85%), driven by Follow-up (56.63%) and Specialist Consultation (55.10%) appointments — not Diagnostic Tests, which actually fall below average (44.00%) in December.
- Patients living more than 20 km from the clinic miss 57.76% of appointments — 11.31 percentage points above the overall average — identifying distance as a logistical barrier.

3.4 Five Business Recommendations

#	Recommendation	Target Segment	Expected Impact
R 1	Cap Follow-up and General booking windows at 14 days	Follow-up + General	~15 pp reduction in these types
R 2	Two-step engagement for High Risk: SMS 3 days before + phone call 24 hours before	High Risk (n=531)	Up to 30 pp improvement vs. no reminder
R 3	Set SMS as the default reminder channel across all patient segments	All patients	4–10 pp improvement over WhatsApp/none
R 4	Flag patients over 20 km for telehealth or transport-assisted options	Distance > 20 km	Reduces logistical barrier for ~11 pp gap
R 5	Increase reminder intensity for November–January seasonal period	Seasonal peak period	Addresses Dec 53.85% and Jan 52.96% spikes

4. Final Validated KPI Results

4.1 KPI 1 — Overall Appointment No-Show Rate

Outcome	Count	Percentage	Operational Impact
No-Show	2,423	48.46%	Empty consulting rooms; wasted capacity
Attended	2,314	46.28%	Completed consultations and patient care
Cancelled	263	5.26%	Advance cancellation; potential rescheduling
Total Bookings	5,000	100.00%	Full dataset baseline

4.2 KPI 2 — Reminder Effectiveness

Reminder Channel	No-Show Rate	vs. No Reminder	Rank
SMS	45.75%	−5.64 pp	1st (Best)
Email	48.41%	−2.98 pp	2nd
WhatsApp	49.77%	−1.62 pp	3rd
No Reminder	51.39%	Baseline	—

4.3 KPI 3 — No-Show Rate by Booking Lead Time

Lead Time Band	No-Show Rate	Appointments (n)	Risk Level
0–7 days (same week)	27.81%	640	Low
8–14 days	33.55%	602	Low-Medium
15–30 days	43.21%	1,333	Medium
31–60 days	60.49%	2,425	High

4.4 KPI 4 — No-Show Rate by Patient Risk Tier

Risk Tier	No-Show Rate	Patients (n)	Definition
Low Risk	43.51%	2,921	0 previous no-shows
Medium Risk	53.49%	1,548	1 previous no-show
High Risk	61.02%	531	2 or more previous no-shows

4.5 KPI 5 — No-Show Rate by Appointment Type

Appointment Type	No-Show Rate	Appointments (n)	vs. Overall
Follow-up	51.23%	1,421	+2.77 pp (Above average)
Diagnostic Test	49.75%	593	+1.29 pp (Above average)
Specialist Consultation	47.44%	900	-1.02 pp (Below average)
General Consultation	46.64%	2,086	-1.82 pp (Below average)

5. Advanced Analytical Findings

5.1 Cross-Variable Analysis: Appointment Type × Lead Time

The table below shows no-show rates at the intersection of appointment type and booking lead time. The Follow-up + 31–60 days combination (65.14%) is the worst-performing segment in the entire dataset.

Appointment Type	0–7 days	8–14 days	15–30 days	31–60 days
Follow-up	29.41%	37.35%	42.94%	65.14% ⚠
Diagnostic Test	29.69%	31.94%	46.15%	59.55%
Specialist Consultation	30.65%	28.30%	44.53%	59.18%
General Consultation	24.60%	33.72%	42.07%	58.10%

5.2 Risk Tier × Reminder Channel

SMS consistently outperforms WhatsApp and Email across all three risk tiers. The Medium Risk + WhatsApp combination (58.59%) is particularly poor — nearly 10 percentage points worse than Medium Risk + SMS (48.64%).

Risk Tier	SMS	Email	WhatsApp	No Reminder
Low Risk	42.52%	43.61%	42.92%	~45%
Medium Risk	48.64%	53.85%	58.59%	~57%
High Risk	56.25%	57.63%	61.36%	68.24%

5.3 Seasonal Patterns

Month	No-Show Rate	vs. Average	Notable Driver
November 2025 (Best)	42.17%	–6.29 pp	Quietest period
April 2025	45.68%	–2.78 pp	Second lowest
Overall Average	48.46%	Baseline	—
July 2025	52.43%	+3.97 pp	Mid-year spike
January 2025	52.96%	+4.50 pp	New-year disruption
December 2025 (Worst)	53.85%	+5.39 pp	Follow-up + Specialist driven

5.4 December 2025 Breakdown by Appointment Type

Appointment Type	December Rate	Overall Rate	Difference
Follow-up	56.63%	51.23%	+5.40 pp ⚠

Appointment Type	December Rate	Overall Rate	Difference
Specialist Consultation	55.10%	47.44%	+7.66 pp Δ
General Consultation	53.40%	46.64%	+6.76 pp
Diagnostic Test	44.00%	49.75%	-5.75 pp $\checkmark$

5.5 Distance Analysis

Distance Band	No-Show Rate	vs. Overall	Implication
Under 5 km	46.45%	-2.01 pp	Lowest barrier
5–10 km	46.80%	-1.66 pp	Near-average
10–20 km	49.43%	+0.97 pp	Slight elevation
Over 20 km	57.76%	+9.30 pp Δ	Significant barrier — telehealth candidate

5.6 Patient Waiting Time (Attended Patients Only)

Metric	Value	Detail
Average waiting time	24.29 min	2,314 attended patients
Maximum waiting time	68 min	Single longest wait recorded
Longest by type	Follow-up: 24.77 min	Also has highest no-show rate
Shortest by type	General: 23.94 min	Also has lowest no-show rate
Busiest day	Tuesday: 25.34 min	Peak demand day
Quietest day	Friday: 22.48 min	Best scheduling efficiency

6. HC-POD Final Cross-Track Integration Record

This section documents the mandatory HC-POD integration record showing how the Data Analytics track connected with other tracks to build the final HealthConnect solution.

6.1 Integration Matrix

Track Relationship	Final Contribution
Data Analytics → Data Science	Validated analytical findings (5 KPIs, 7-variable feature importance, extreme segments) supported final model feature engineering and evaluation
Data Analytics → Project Management	Final business insights, 5 recommendations, and presentation-ready KPI results provided for final project documentation
Data Science → Data Analytics	Model feature importance confirmation (booking_lead_days #1, previous_no_shows #2, reminder_sent #3) validated the analytical framework direction
All Tracks → HealthConnect Solution	Each track component connects to answer: "How can HealthConnect use data and AI to reduce missed appointments?"

6.2 Detailed Integration Record: Data Analytics ↔ Data Science

Integration Item	Detail
Track collaborated with	Data Science Track
Dependency	Data Science required validated analytical patterns to select and engineer features for the predictive no-show model
Output received	Confirmation that lead_time_band and patient_risk_tier as categorical engineered features (from Data Analytics) improved model precision and recall. Feature importance ranking confirmed: booking_lead_days (#1), previous_no_shows (#2), reminder_sent (#3)
Output provided	Feature importance table (7 variables), cross-variable charts, worst-case segment (72.31%), best-case segment (23.18%), High Risk intervention benchmark (37.93%), December appointment type breakdown
Final integration activity	37.93% High Risk best-case rate adopted as the model evaluation benchmark — the minimum performance threshold the Data Science model must beat to demonstrate improvement over analytics-only intervention
What changed	Data Science model moved from using raw continuous variables to validated categorical bands (lead_time_band, patient_risk_tier), which improved both model interpretability and predictive performance
How the change improved the solution	The final HealthConnect solution now has an analytically-grounded model evaluation standard. The 37.93% benchmark ensures the AI component adds measurable value beyond what targeted SMS reminders alone can achieve for High Risk patients

Integration Item	Detail
Evidence	Week 7 Tests T1–T8 (all passed). Feature importance table produced in Week 6 notebook. High Risk benchmark calculation in Week 7 Refinement R2 (n=29, 37.93%).
Contribution to presentation	Data Analytics will present the 49.13 pp segment gap (23.18% best vs. 72.31% worst) as the business case for the integrated solution, and the 37.93% benchmark as evidence of the model's minimum required value

7. Final End-to-End Integration Walkthrough

This section describes how the Data Analytics component fits into the complete HealthConnect solution.

Walkthrough Step	Data Analytics Contribution
1. The Problem	HealthConnect Clinic's 48.46% no-show rate represents nearly half of all booked appointments going unfulfilled — the central operational problem the entire project addresses
2. The Data	5,000 appointment records across 18 variables, January 2025–June 2026. Data quality audit (Week 4), cleaning and feature engineering (Week 5) produced a validated analytical foundation
3. The Analytics	5 KPIs revealed: lead time dominates outcomes (27.81% vs. 60.49%), High Risk patients need two-step intervention, SMS is the optimal channel, and December/January are peak risk months
4. The Predictive Component	Validated patterns and engineered features (lead_time_band, patient_risk_tier) provided to Data Science. The 37.93% High Risk best-case rate set the model evaluation benchmark
5. The ML Engineering Component	Data Analytics confirmed required input variables: booking_lead_days, previous_no_shows, reminder_sent, appointment_type — standardised and validated for pipeline ingestion
6. The GenAI Component	No-show risk profiles by segment provided context for the AI administrative assistant to recommend appointment timing and reminder strategies to clinic staff
7. The Project Management Layer	Business recommendations (R1–R5) and quantified insights provided to PM for final presentation and stakeholder briefing materials
8. The Final Solution	Analytics + Predictive Model + AI Assistant + ML Pipeline = an integrated system that identifies high-risk appointments in advance and triggers the right intervention at the right time
9. Business Value	If the best-case conditions (SMS + same-week booking) were applied to the worst segment (High Risk + 31–60 days), the rate could drop from 72.31% to ~37.93% — a potential 34-point reduction for the highest-risk patients
10. Limitations	Findings based on historical data only. No qualitative reason data for cancellations. Single clinic dataset. Waiting time unavailable for no-shows. Generalisation requires validation on new data.

8. Final Analytics Dashboard

The HealthConnect analytics dashboard is an interactive five-page HTML report presenting all validated KPI results. It was built using Python-generated data and HTML/CSS/JavaScript, and is accessible via any web browser without additional software.

Dashboard Page	Title	Primary KPI
Page 1	Executive Overview	KPI 1 — Overall No-Show Rate, segment cards
Page 2	No-Show Patterns	KPI 5 — No-Show by Appointment Type, day, time, age group
Page 3	Reminders Analysis	KPI 2 — Reminder Effectiveness and channel comparison
Page 4	Booking Lead Time	KPI 3 — No-Show by Lead Time Band, 18-month trend
Page 5	Patient Risk	KPI 4 — No-Show by Risk Tier and prior history

Dashboard file: HealthConnect_Dashboard.html — open in any modern browser. All values displayed match the Week 7 validated results confirmed by Tests T1–T8.

9. Project Evolution — Week 4 to Week 8

Week	Focus	Key Output	Status
Week 4	Problem Understanding	KPI proposal, data quality audit, 5,000-record overview	✓Complete
Week 5	EDA & KPI Calculations	All 5 KPIs calculated, 12-dimension EDA, 5 recommendations	✓Complete
Week 6	Advanced Analytics & Integration	Cross-variable analysis, feature importance table, Data Science integration	✓Complete
Week 7	Testing & Refinement	8 formal tests (all pass), 3 refinements, 5-page dashboard	✓Complete
Week 8	Final Integration & Presentation	Executive summary, integration record, presentation package	✓Complete

10. Analytical Limitations

Limitation	Detail	Impact
No qualitative cancellation data	The dataset does not record why patients missed — no financial, transport, or emergency reason data	Recommendations are behavioural/logistical only; root causes may differ
Single clinic dataset	5,000 records from one fictional clinic across 18 months	Findings may not generalise to larger networks without re-validation
Waiting time gap	waiting_time_minutes only available for attended patients (n=2,314)	Cannot use waiting time to predict no-shows; operational metric only
Historical assumption	Analysis assumes past booking behaviour predicts future attendance	Intervention effectiveness must be measured prospectively
Reminder channel self-selection	Patients chose their channel; SMS patients may inherently differ from WhatsApp patients	Channel effect and patient-type effect cannot be fully separated
Distance imputation	90 records (1.8%) had distance imputed using median	Minor potential bias in distance-band analysis

11. Professional Reflection

11.1 Key Takeaway

The most important lesson from this project is that aggregate averages hide actionable insights. The overall no-show rate of 48.46% looks like a single problem — but when disaggregated by lead time, risk tier, reminder channel, and appointment type simultaneously, a 49-percentage-point gap emerges between the best and worst patient segments. Data analytics does not just describe the problem: it makes the right intervention visible.

11.2 What I Learned

- Data validation is not optional. The Week 7 testing phase confirmed that investing time in formal tests (T1–T8) builds the kind of confidence needed to present findings to stakeholders without qualification.
- Feature engineering matters. Transforming `booking_lead_days` into four categorical bands (`lead_time_band`) and `previous_no_shows` into three risk tiers improved both analytical clarity and the Data Science model's performance — the same insight, applied twice across tracks.
- Cross-track integration requires actual exchange. The most valuable integration outcome was the 37.93% High Risk benchmark — a number that did not exist until the Data Analytics track ran Refinement R2 and shared it with Data Science. Communication alone is not integration.
- Business framing is as important as technical accuracy. A no-show rate of 72.31% for High Risk + 31–60 day patients means very little until it is placed alongside 37.93% under the right conditions — the gap is the business case.

11.3 Future Improvements

- Add qualitative reason capture to the appointment booking form to separate logistical barriers from motivational ones.
- Test a live A/B reminder experiment across SMS, WhatsApp, and two-step protocols to move from correlational to causal evidence.
- Extend the dataset across multiple clinic sites to validate whether the lead-time and risk-tier patterns hold at scale.
- Integrate the predictive model output into the booking system to automatically flag high-risk appointments at the point of scheduling.

12. Final Submission File Inventory

File	Description	Week
Week4_HealthConnect_DataAnalytics.ipynb	Problem understanding, KPI proposal, data quality audit	W4
Week4_HealthConnect_SQL_Queries.sql	Baseline SQL validation queries	W4
Week4_DataAnalytics_HealthConnect.docx	Week 4 project report	W4
Week5_HealthConnect_DataAnalytics.ipynb	EDA, KPI calculations, visualisations	W5
Week5_ProjectSummaryReport_HealthConnect.docx	Week 5 report with all KPI results	W5
Week6_HealthConnect_DataAnalytics.ipynb	Advanced analytics, cross-variable analysis	W6
Week6_ProjectReport_HealthConnect.docx	Week 6 report including integration record	W6
Week7_HealthConnect_DataAnalytics.ipynb	8 formal tests + 3 refinements	W7
Week7_ProjectReport_HealthConnect.docx	Week 7 testing and refinement report	W7
HealthConnect_Dashboard.html	Interactive 5-page analytics dashboard	W7
Week8_HealthConnect_FinalReport.docx	This document — final integration portfolio	W8
README.md	GitHub repository final README	W8